

Euclidean Geometry In Mathematical Olympiads

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Problem 1.51 (IMO 1985/1)

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2. Notice that there are two configurations which one of them shows T between A and O and the other one shows T between B and O , it might be better to use the **directed angle** instead of the **indirected angle**.

Tafi_ak

311 posts

Jan 11, 2023, 5:02 AM

#30

Let ω be that circle which is tangent to the three sides. If $AD \parallel BC$ then AD, BC have to be tangent and hence obviously the radius is equal to AD and BC . So we are done for this case.

Let AD and BC are not parallel and $E = AD \cap BC$, wlog $EA < EB$. Let I be the incenter of $\triangle EDC$. Let $J = AI \cap (EDC)$, $K = AI \cap AB$. The center of ω lies on AB means K is actually the E -excenter of $\triangle EAB$. Notice that $\angle CBK = \angle EDC = \angle EJC$, means $CBKJ$ is cyclic. Similarly $DAKJ$ is cyclic.

By fact 5, we have $KIDC$ is cyclic with center J . So $JD = JK$ means AJ is the angle bisector of $\angle DAK$. Therefore J is the incenter of $\triangle EAB$.

Let $X = AB \cap (KIDC) \neq K$. Notice that $DAXJ$ and $CBXJ$ are a kite. So $DA = AX, BC = BX$. Hence $AD + BC = AB$.

cinnamon_e

704 posts

May 25, 2023, 8:21 PM

#31

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Let T be a point on AB such that $AD = AT$, and O be the center of the circle centered on AB . Note that

$$\angle DCO = \frac{1}{2}\angle DCB = \frac{1}{2}(180^\circ - \angle DAB) = \frac{1}{2}(180^\circ - \angle DAT) = \angle ATD,$$

so $\angle OTD + \angle OCD = 180^\circ$. Hence, T lies on (OCD) .

Now,

$$\angle BTC = \angle OTC = \angle ODC = \frac{1}{2}\angle ADC = \frac{1}{2}(180^\circ - \angle ABC) = \frac{1}{2}(180^\circ - \angle TBC),$$

so $\angle BTC = \angle BCT$.

Therefore, $BC = BT$, so $AD + BC = AT + BT = AB$, as desired.